

Introduction to Cloud Computing

Assignment 1

Building a Basic Web App Using Azure App Service

Objective: In this assignment, you will create a simple web application hosted on Azure App Service. This project will help you gain hands-on experience with Azure App Service, focusing on web deployment, configuration, and management without using any additional Azure services.

Assignment Steps:

1. Setup and Configuration of Azure App Service

- **Task:**
 - Create a new App Service instance on Azure. The name must contain your Project Group Code and your Student Id.
 - If I have Project Group D, and my Id is K22-1234 I will name the website “**d221234**” in my appservice.
 - Configure the App Service settings, including selecting the correct runtime stack (C# Dot Net Latest) and region for optimal performance.

Deliverable:

- A screenshot of your Azure App Service configuration page showing the app service details.
- A 200-word explanation of your choices for the runtime stack, pricing tier, and region.

2. Development of a Basic Web Application

- **Task:**
 - Develop a simple static or dynamic web application using C# Dot Net Latest. The app should have at least:
 - A page showing Introduction of your Project along with your Group details (i.e. GroupCode, Member, Leader etc) and your brief project plan.
 - A page showing and discussing the architecture of your semester project.
 - A page discussing azure services which will be used in your semester project.

Deliverable:

- Source code of your web application (zipped and uploaded to the Classroom).

- A brief description (200 words) of the app's functionality.

3. Deploy the Web App to Azure App Service

- **Task:**

- Deploy your web application to the App Service you created.
- Ensure the application is live and accessible via a public URL.
- The URL must contain your name and id.

Deliverable:

- Live URL of the web application hosted on Azure App Service.
- Screenshot of the App Service dashboard showing the deployed app.
- A short description (200 words) of any challenges you encountered during the deployment and how you resolved them.

Deliverables Checklist:

1. **App Service Setup and Configuration** – Screenshot and 200-word explanation of the App Service configuration.
2. **Basic Web Application** – Source code of the web application and 200-word description of functionality.
3. **Deployment** – Live URL of the deployed web app, deployment screenshots, and 200-word description of deployment challenges.

You must place the source code inside a zip and a single PDF document (not in zip) containing all Screenshots, URLs and other details required

Grading Rubric:

- **App Service Setup (15 points):** Proper configuration and explanation of App Service.
- **Web Application Development (15 points):** Functional and user-friendly web application.
- **Deployment (20 points):** Successful deployment with working URL, and clear handling of deployment issues.

Submission Deadline:

- All deliverables must be submitted by **5th October 2025, 11.59PM**.

Notes:

- It is an **individual** activity. Your web pages should not be visually similar to anyone else, but you can help each other with service creation and deployments.
- The web app should remain live and accessible for review until grading is completed.
- You must use Azure App Service exclusively for hosting and management. No other Azure services (e.g., databases, storage, etc.) are allowed.